
Sustainable Development of Green Communication through Threshold Visual Cryptography Schemes Using a Population Based Incremental Learning Algorithm

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Abstract

Today, in the era of the Green Internet, information security is an extremely important entity. As a consequence of sending some sensitive information to the network using common channels of communication, this has become completely inevitable. There is a specific application of Visual Cryptography (VC) to send images since either two or more VCs having equal black and color pixels are present. This secret message may be decrypted by means of superimposing VC shares along with the Human Visual System (HVS) that discloses a secret. The results of such pixel expansion in the VC the decrypted image can be reduced. The VC also lacks such general approaches in constructing schemes of Visual Secret Sharing (VSS) for the structures of general access. This is further enhanced by subjecting the VC shares to the techniques such as Population-Based Incremental Learning (PBIL) or Particle Swarm Optimization (PSO) that are nature-inspired. This also improves the sharpness and quality of an image. The results of the experiment proved that the method performed better than the other methods.

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1 Introduction

Cryptography, in today's world, is fast becoming a necessity for all types of organizations. It refers to a technique that is employed to avoid any unauthorized data access. There are two major components to cryptography. They are a) Encryption algorithm, and b) Key. At times, there are multiple keys used for encryption. For cryptography, certain crucial terms are provided below: Plaintext refers to the original text that is encrypted, and ciphertext is the text that is encrypted. The text will obtain this after data encoding using a key called the ciphertext. Key is the value used for encrypting a plain text or a decrypt of the ciphertext. Encryption – this is the method that converts data into a coded form using the key. Decryption – this refers to the actual method that converts encoded data to its original form. Confidentiality and authentication are the two salient features of cryptography. Confidentiality refers to the protection received against any unauthorized information disclosure. It may further be applied to the entire messages, parts of certain messages, and the existence of such messages. Confidentiality refers to the protection of all transmitted data obtained from passive attacks. Authentication services are for ensuring the authenticity of communication [1][22].

It may be a challenging task to ensure information security as there can be problems faced at the time of information sharing. Also, sharing of any type of unauthorized contents of multimedia in an illegal manner and copying of data for misuse can be common problems with regard to security. Generally, cryptography can play a crucial role in improving the security level of information. This cryptographic approach will work based on numerical computation and is known as key cryptography. Identifying this more significantly, using of various keys that perform decryption, as well as encryption can be called the public key cryptographic method known as the Elliptic Curve Cryptography (ECC) and this will work in a similar manner to the method of public key cryptography and this essentially means that the ECC will have various keys for decryption and encryption. In the ECC, even before plaintext encoding is done and as soon as decoding is done, the original translation of data will be performed. But one single spot on the curve is encrypted or decrypted only by using the ECC and not their messages. Furthermore, the ECC can generate a public or a private key in a random manner thus increasing the needs for computation and time taken for processing while securing any image. Thus, it becomes important to develop

an encryption algorithm that is efficient in order to increase the confidentiality and privacy of its original image [2].

Visual Cryptography (VC) had been suggested by Naor and Shamir, permitting secret information encryption in the form of an image. By means of applying this secret-sharing concept, there is a secret image that is encrypted and other images are shared and printed on transparencies that can be further distributed to the n participants. By means of stacking n transparencies (shares) in a direct manner, these secret images may be exposed and also recognized visually by humans who do not have any cryptographic knowledge or computational devices. At the same time, either a single share or a small portion of shares cannot leak anything that may be related to a secret image. The VC can be a great solution to share secrets at the time computers are not always employed for the process of decryption [3].

In the recent decade, there have been several results on research made on the Visual Secret Sharing (VSS) scheme (that is called k out of n VSS scheme or $((k, n) - \text{VSS})$ scheme) that were proposed. There are conventional VSS schemes generating noise-like pixels on the shares for hiding secret images. This way, a secret may be concealed perfectly on the share images. But these schemes tend to suffer from problems of management where the dealers are incapable of identifying every share visually. The problem of pixel expansion can be a very common disadvantage of having many VSS schemes. At the time of applying the VC-based approach, every secret pixel in the secret image can be encrypted within a block that contains m sub-pixels for every constituent share image [4]. Therefore, the entire area of this share will be m times of the original secret image. Any contrast of such recovered images is decreased to $1/m$ at the same time. The problem of pixel expansion affects the needs of both transmission and storage for the shares but can also decrease the level of contrast of the secret images that have been recovered.

Aside from the drawbacks mentioned above, all the constructions made for different threshold VC schemes were designed in a specific manner. These are the issues that have been motivating us to be able to develop another systematic approach for the construction of these general threshold VC schemes for dealing with the issues of pixel expansion and improving the quality of the display of these recovered images. The problem of construction of pixel-expansion-freed (k, n) -VC schemes that are on the binary secret images in the form of a mathematical model of optimization is made first.

In the case of a standard point of view, this process of optimization [5] may be described as the one that identifies an ideal solution of a function from a system that may have constraints. The process will have to follow certain components which are as follows: The result of this function will need either a minimized or maximized process which is for processing metrics that are based on the cost, profit, and yield of the system. This will convert the issue of optimization within the group consisting of equations along with inequalities in the order of constraints. Such constraints were used

for comprising of an area that defines all limitations in the process of this system. The variables of the predictive model will have to fulfil constraints. This is achieved by several instances consisting of variable values that guide it to an area established by the subspace of such variables. Such subspaces were classified by means of a group of decisions in various engineering issues that was understood as a level of freedom in this process. The algorithm that is chosen will ensure the system can give the best solution to the problem. There are some traditional and important algorithms that are explained as given: PSO algorithm, Differential Evaluation (DE) algorithm, Ant Colony Optimization (ACO), Cuckoo Search (CS) Algorithm, and the Genetic Algorithm (GA).

The main goal of the model [6] for maximizing the recovered images that were subject to security and blackness constraints. The proposed method was a systematic approach applicable to the (k, n) –VC schemes for both general n and k . This means it may not be needed for the design of the (k, n) –VC schemes that were in the earlier Visual Cryptography Scheme (VCS) methods of construction. This further develops an effective algorithm based on the PBIL and PSO approach in solving such complicated and challenging problems. In accordance with the results of computation, this had evaluated the performance that verifies the effectiveness of these images. The remainder of the investigation has been organized thus. The related work in literature is discussed in Section 2. The different methods used are shown in Section 3. The experimental results are discussed in Section 4 and the conclusion is made in Section 5.

2 Literature Survey

Tagged Visual Cryptography Scheme (TVCS) has added tag images to noise-like shares that were generated by the VCS for improving shares management of a traditional VCS. But, the currently existing TVCS will suffer visual quality for the secret image that is degraded resulting in pixel expansion. Chiu and Lee [7] had proposed another Threshold Adaptive Tagged Visual Cryptography Scheme (k, n) –ATVCS in solving the problems mentioned above. This problem of ATVCS encryption was formulated for a model of mathematical optimization which was an evolutionary algorithm that has been developed in finding an optimal solution. The (k, n) –ATVCS proposed has enabled an encryptor in adjusting the quality between a tag image tuning with a secret image using tuning parameters. The results of the experiment proved the effectiveness and correctness of this work.

Shivani and Agarwal [8] had proposed another novel and verifiable VC approach that was progressive using the additional embedding capacity for every share. This also included self-embedding authentication data, confidential payload, and copyright information. All embedded information was retrieved with a confidential payload during the conflict. This approach was also able to eliminate unwanted constraints of encryption such as pixel expansion, noise-like shares, and the need for a codebook. The experiments

proved that although there are participants with various credentials that were embedded in shares, the contrast was 50 % without any reduction to the secrecy level. It had been confirmed through experiments that the approach was able to localize the share's tampered region effectively.

Kaur and Singh [9] proposed another algorithm to preserve image security from the Principal Component Analysis (PCA) along with Local Pixel Grouping (LPG). The technique helps preserve the local structure of the image. The first iteration in this was to add to the embedding method and the second one was to add to the process of message extraction. This LPG-PCA algorithm helps in providing better results than the Wavelet Transformation (WT). Various experiments proved the effectiveness of this algorithm with the proposed metrics. A basic disadvantage of the VC was pixel expansion wherein each pixel was substituted using m sub-pixels for each share resulting in resolution loss. Therefore, enhancing the visual nature of the VC was researched very often. Mary and Rani [10] had proposed another technique to improve the VC and its visual quality by using the Ant Colony Optimization (ACO) considering a varied range of images that were in color and grey. This technique helped in building the sharpness and the quality of the image. It subjectively assessed the perception of human vision with standard measurements.

In the case of traditional systems, data are encrypted using the key but which chances of being eavesdropped. Thus, the key will have to be unpredictable and strong and therefore a method was proposed to take advantage of the natural selection theory. Nazeer et al., [11] had further proposed the GA to be a tool that solves various problems by means of modelling certain genetic processes that were simplified and were considered to be of another class. By means of employing the GA, the key and its strength improved and made the entire algorithm better. For this method proposed, the data are encrypted using various steps. Firstly, the key is generated using a random number generator and by the application of genetic operations. After this, data are diffused using genetic operators and then logical operators were performed in the diffused data which served as the key for data encryption. Finally, there was a comparative study that was made between the method proposed and the other cryptographic algorithms. It was also observed that the algorithm proposed showed better results in terms of the strength of the key but was computationally less efficient compared to the other two.

Gupta and Jain [12] had proposed a system that highlights a more novel approach creating a method of steganography that makes use of the VC and the GA for robust encryption of computer forensics. Even though extensive work or research has been made in the past, most of such research work does not have any optimal consideration for robust security made towards an encrypted image. The methods proposed can encode a secret message that is given in small insignificant bits of their original image wherein pixel values of an encrypted image can be modified by using the GA. This is to retain all

their statistic traits thus making secret message detection a challenge. Using the GA has been compelling the system to enhance security by making use of mutation, crossover, and optimal selection. This system can hide data within the real image to achieve detection once it underwent visual cryptography. An implementation is made on the java platform showing the proposed system to have better levels of resilience by taking into consideration Steganalys is along with benchmarking them with visual standards that are optimal.

In order to make use of Vector Quantization (VQ), the two parameters that are the width of the cell and the dimension of the grid will have to be optimized. Das et al., [13] had proposed the PSO-guided VQ which was a multilayer perceptron in conjunction with the auto-encoder trained in line with it to automatically get an optimal VQ for every type of image at the time of transmission. This scheme had implemented various images successfully for a secure transmission using a 62.8% reduction in data volume and 98.07% retrieval of image quality. The VC is a very strong method in cryptography. The primary advantage of the system was that decryption does not require any specific form of decoding except the human eye. By making use of the half-toning techniques all binary images were obtained and the technique was applied to Halftone VC. Thomas and Gharge [14] had implemented the Direct Binary Search (DBS) with modified images for better recovery of images. This concept was proposed for the bank lockers. There was a comparison made while making use of such parameters like the Peak Signal-to-Noise Ratio (PSNR), Structural Similarity (SSIM), Correlation, and Universal Quality Index (UQI). The VC can split visual content into n image components and such components were further sent to the network. The receiver then superimposes the components of the image to obtain its original information. Kumari et al., [15] introduced an improvement to the security of the VC Blowfish algorithm. using this algorithm every share that was generated by the VC was encrypted with a secret key. On the receiver's side, the receiver will have to decrypt the image components and also overlap them to obtain an original image. By making use of this method, the issue of security of the current system of the VC was mitigated.

Shankar et al., [16] had made a proposal of a wavelet-based secret image scheme for sharing using encrypted shadow images and techniques of optimal Homomorphic Encryption (HE). The Discrete Wavelet Transform (DWT) was initially applied to the secret image for producing sub-bands. From the process, there were multiple shadows created, encrypted, and finally decrypted for every shadow. This encrypted shadow was recovered by selected a subset of the ' n ' shadows to make them transparent and to stack them on each other. For improving the security of the shadow, every shadow will be encrypted and then decrypted by using HE technique. For the problems related to the quality of the image, an Oppositional-based Harmony Search (OHS) algorithm had been used for the generation of an optimal key. From this analysis, it was evident that the scheme proposed was able to provide better levels of security in comparison with the other schemes in existence.

3 Methodology

For the purpose of this work, another method based on optimization for the (k, n) –VSSs was proposed for encryption of binary secret images. The process of decryption can directly stack the shared images printed on transparencies. Here, the PBIL and the PSO are discussed.

3.1 Particle Swarm Optimization (PSO) Algorithm

The PSO refers to a population-based stochastic technique of optimization developed in the year 1995 by Dr. Eberhart and Dr. Kennedy. This was inspired by the social behavior of fish schooling or bird flocking. It shares several similarities with other techniques of evolutionary computation like the GA. This system was initialized using a random solution population and looks for generations of optima. In the PSO, there are some potential solutions called particles that fly through a problem space that follows the current optimum particles [17]. This is initialized using a group of random particles to search for optima. This is done by updating generations. For each iteration, the particle will be updated using the two "best" values. The first is the best fitness solution achieved until now called the *pbest*. The next is the best value tracked by the optimization that is the best one obtained until now by any particle. This is known as the *gbest* or the global best. If a particle participates in the population as a topological neighbor the value is called *lbest* or the local best. Once both these best values [18] are found, the velocity and the positions are updated by the particle using equations (1) and (2).

$$v_i^{t+1} = W^t \cdot v_i^t + c_1 \cdot r_1 \cdot (pbest_i^t - X_i^t) + c_2 \cdot r_2 \cdot (gbest^t - X_i^t) \quad (1)$$

$$X_i^{t+1} = X_i^t + v_i^{t+1} \quad (2)$$

Wherein, X_i^t and v_i^t indicate the position and the velocity of that of the i^{th} particle at a time t , W^t denotes the inertia weight at t^{th} time instant. c_1 and c_2 are the positive acceleration constants and r_1 and r_2 denote random values [0, 1], in a uniform distribution. $Pbest_i$ refers to the best solution in the i^{th} individual particle over a flight path. $Gbest$ refers to the best particle in the entire generation until now.

According to Equation (1), the earlier velocity v_i^t refers to the inertia component that can remember the direction of the earlier flight that can prevent this particle from changing direction. The cognitive component which is $c_1 \cdot r_1 \cdot (pbest_i^t - X_i^t)$ quantifies a performance that is relative to the past performances and this can remember the previous best position. The Social component $c_2 \cdot r_2 \cdot (gbest^t - X_i^t)$ will quantify the performances that are relative to neighbors. To start the algorithm, all particle positions are initialized randomly and their velocities are set to 0 or smaller random

values. The Particle dimension (n_x), Swarm size (n_s), Inertia weight (W), Acceleration coefficients (c_1 and c_2), and the number of iterations (n_t), are initialized .

For all generations, every particle will be accelerated to the previous best position or the global best position. For every such new particle, the velocity will be computed based on the current one along with the distance from the earlier best and the global best. The subsequent position of the particle within the search space will be computed on the basis of the new velocity. This process is iterated about n_t number of times or until the minimum error has been met. This algorithm is then terminated as soon as the particle fitness has been achieved to its desired value or until the n_t iterations are complete. Pseudocode for the PSO is given in Algorithm 1 [19]:

Algorithm 1: Pseudo code for a PSO Algorithm

```

Begin;
  Generate random population of  $N$  solutions (particles);
  For each individual  $i \in N$ : calculate fitness ( $i$ );
    Initialize the value of the weight factor,  $w$ ;
    For each particle;
      Set  $pBest$  as the best position of particle  $i$ ;
      If fitness ( $i$ ) is better than  $pBest$ ;
         $pBest(i) = \text{fitness}(i)$ ;
    End;
    Set  $gBest$  as the best fitness of all particles;
    For each particle;
      Calculate particle velocity according to Eq. (1);
      Update particle position according to Eq. (2);
    End;
    Update the value of the weight factor,  $u$ ;
    Check if termination=true;
  End;
End

```

3.2 Proposed Population Based Incremental Learning (PBIL) Algorithm

PBIL was initially proposed by Baluja, and was a simple form of Estimation of Distribution Algorithms (EDAs). It applies a probability vector that represents the population and every element refers to the probability that obtains a value 1 for a particular gene position. This probability vector was updated to choose the best solution using the Hebbian learning rule. The

PBIL has proven to be very successful on different benchmark functions compared to the GA and is also used for real-world problems [20]. In the PBIL, the individual population is denoted by the probability vector as shown in (3):

$$p_l(x) = (p_l(x_1), p_l(x_2), \dots, p_l(x_n)) \quad (3)$$

Wherein, $p_l(x_i)$ is the probability of the value 1 found in the i^{th} gene position for the l^{th} generation.

This algorithm will begin by means of initialization of its probability vector $p_0(x)$ along with the elements set to 0.5. This ensures that at the time of sampling from the probability, the vector random solutions were created owing to the probability of generating it as 1 or 0 which is equal. With the progress of the search, the values in the probability vector were learned towards the values that represent solutions with high evaluations. The PBIL process of evaluation will be accomplished as given below: For each generation, the M solutions will be generated based on the probabilities mentioned in their current probability vector $p_1(x)$. These solutions are further evaluated with the N best solutions ($N \leq M$) that are selected. This is denoted by: $x_{1:M}^l, \dots, x_{i:M}^l, \dots, x_{N:M}^l$. The best solutions chosen were used for updating a probability vector using the Hebbian inspired rule as in (4):

$$p_{l+1}(x) = (1 - a)p_l(x) + a \frac{1}{N} \sum_{K=1}^N x_{k:M}^l \quad (4)$$

Wherein, $a \in (0,1]$ refers to the learning rate.

Once the probability vector has been updated, another new solution set will be generated by means of sampling from a new probability vector $p_{l+1}(x)$. This cycle will be repeated until such time the termination condition is met. This may be like the probability vector being converged to 0.0 or 1.0 for every bit position.

The pseudocode for the PBIL is given Algorithm 2 [21]:

Algorithm 2: Pseudo code for a PBIL Algorithm

Begin;

Initialize probability vector $p_0(x)$

While no convergence do

Begin

Using $p_1(x)$ obtain M solutions: $x_1^l, \dots, x_M^l$

Evaluate and rank $x_1^l, \dots, x_M^l$

Select the $N(N \leq M)$ best solutions: $x_{1:M}^l, \dots, x_{N:M}^l$

Update the probability vector

$p_{l+1}(x) = p_{l+1}(x_1), \dots, p_{l+1}(x_n)$

For $i=1 \dots n$ do

$p_{l+1}(x) = (1 - a)p_l(x) + a \frac{1}{N} \sum_{K=1}^N x_{k:M}^l$

End

4 Results and Discussions

In this section, the PSO and PBIL methods are used. Experiments are carried out using (k, n) and $k = 5, n$ for different blackness range (100 to 70). The contrast (various blackness range) as shown in tables 1 to 6 and figures 1 to 6.

Table 1 Contrast for PBIL

(k, n) for different Blackness Range (100,90)	PSO	PBIL
(2,3)	35.1	36
(2,4)	30.9	31.7
(2,5)	30.2	31
(2,6)	29.2	30
(2,7)	27.2	27.9
(2,8)	26.8	27.4

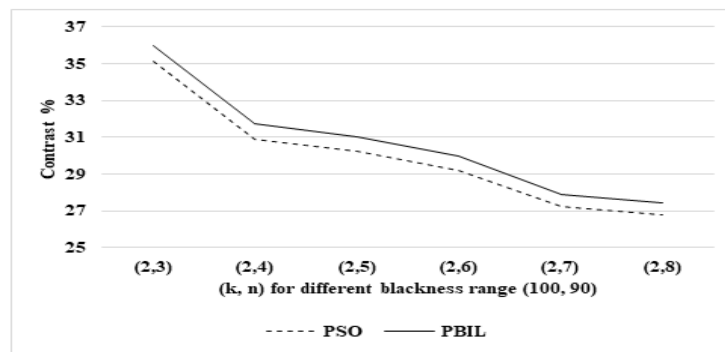


Figure 1 Contrast for PBIL

From the figure 1, it can be observed that the PBIL has higher contrast by 2.53% for (2, 3) blackness range (100, 90), by 2.55% for (2, 4) blackness range (100, 90), by 2.61% for (2, 5) blackness range (100, 90), by 2.7% for (2, 6) blackness range (100, 90), by 2.54% for (2, 7) blackness range (100, 90), and by 2.21% for (2, 8) blackness range (100, 90) when compared with PSO respectively.

Table 2 Contrast for PBIL

(k, n) for different Blackness Range (90,80)	PSO	PBIL
(2,3)	36.3	37.1
(2,4)	32	32.8
(2,5)	31.3	32.2
(2,6)	30.4	31.2
(2,7)	28.1	28.8
(2,8)	27.8	28.6

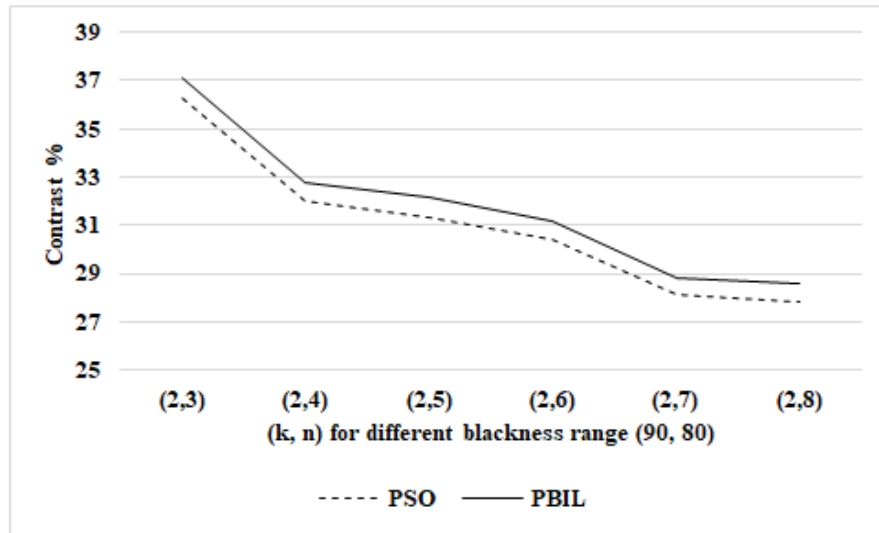


Figure 2 Contrast for PBIL

From the figure 2, it can be observed that the PBIL has higher contrast by 2.18% for (2, 3) blackness range (90, 80), by 2.47% for (2, 4) blackness range (90, 80), by 2.83% for (2, 5) blackness range (90, 80), by 2.59% for (2, 6) blackness range (90, 80), by 2.46% for (2, 7) blackness range (90, 80), and by 2.83% for (2, 8) blackness range (90, 80) when compared with PSO respectively.

Table 3 Contrast for PBIL

(k, n) for different Blackness Range (80,70)	PSO	PBIL
(2,3)	37.5	38.4
(2,4)	33.1	34.1
(2,5)	32.1	32.8
(2,6)	31.3	32.1
(2,7)	28.9	29.7
(2,8)	28.7	29.5

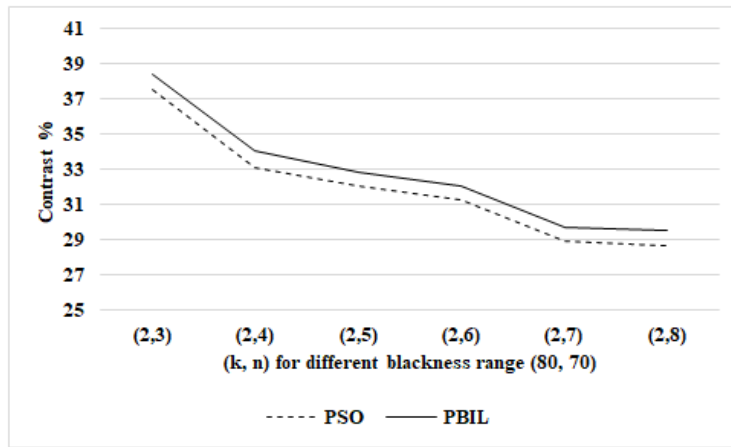


Figure 3 Contrast for PBIL

From the figure 3, it can be observed that the PBIL has higher contrast by 2.37% for (2, 3) blackness range (80, 70), by 2.97% for (2, 4) blackness range (80, 70), by 2.15% for (2, 5) blackness range (80, 70), by 2.52% for (2, 6) blackness range (80, 70), by 2.73% for (2, 7) blackness range (80, 70), and by 2.75% for (2, 8) blackness range (80, 70) when compared with PSO respectively.

Table 4 Contrast for PBIL

$(k = 5, n)$ for different Blackness Range (100,90)	PSO	PBIL
(2,5)	30.2	31
(3,5)	20.2	21.6
(4,5)	14.2	14.3

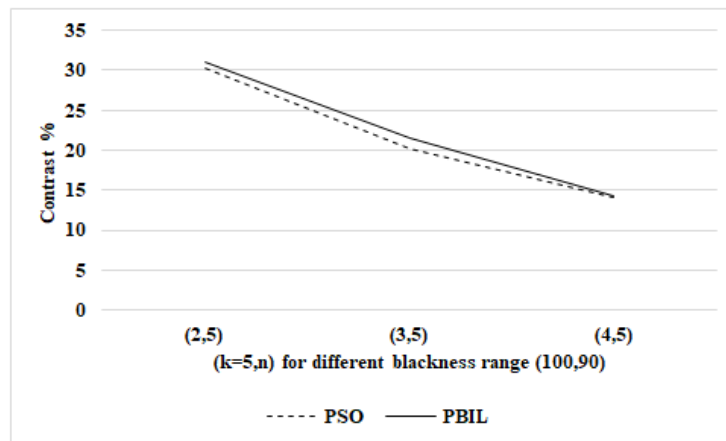


Figure 4 Contrast for PBIL

From the figure 4, it can be observed that the PBIL has higher contrast by 2.61% for (2, 5) blackness range (100, 90), by 6.69% for (3, 5) blackness range (100, 90) and by 0.7% for (4, 5) blackness range (100, 90) when compared with PSO respectively.

Table 5 Contrast for PBIL

$(k = 5, n)$ for different Blackness Range (90,80)	PSO	PBIL
(2,5)	31.3	32.2
(3,5)	21.6	21.6
(4,5)	15	14.2

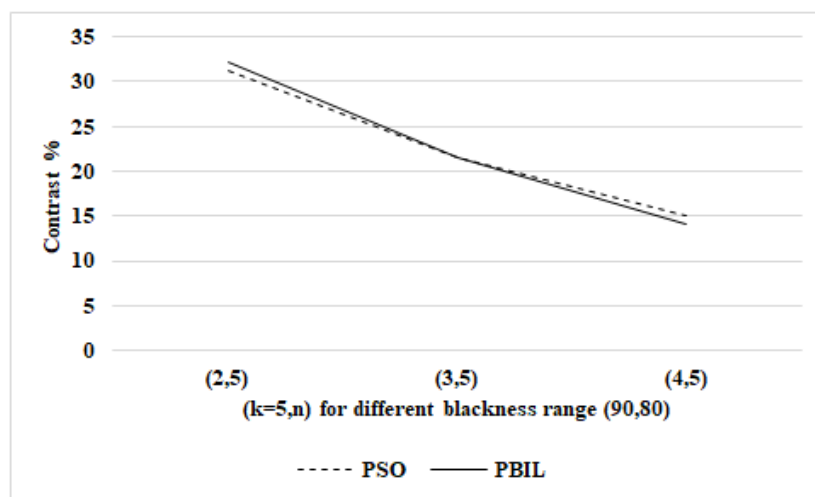


Figure 5 Contrast for PBIL

From the figure 5, it can be observed that the PBIL has higher contrast by 2.83% for (2, 5) blackness range (90, 80), by no change for (3, 5) blackness range (90, 80) and by 5.48% for (4, 5) blackness range (90, 80) when compared with PSO respectively.

Table 6 Contrast for PBIL

$(k = 5, n)$ for different Blackness Range (80,70)	PSO	PBIL
(2,5)	32.1	32.8
(3,5)	21.5	22.9
(4,5)	14.8	15.7

From the figure 6, it can be observed that the PBIL has higher contrast by 2.16% for (2, 5) blackness range (80, 70), by 6.31% for (3, 5) blackness range (80, 70) and by 5.9% for (4, 5) blackness range (80, 70) when compared with PSO respectively

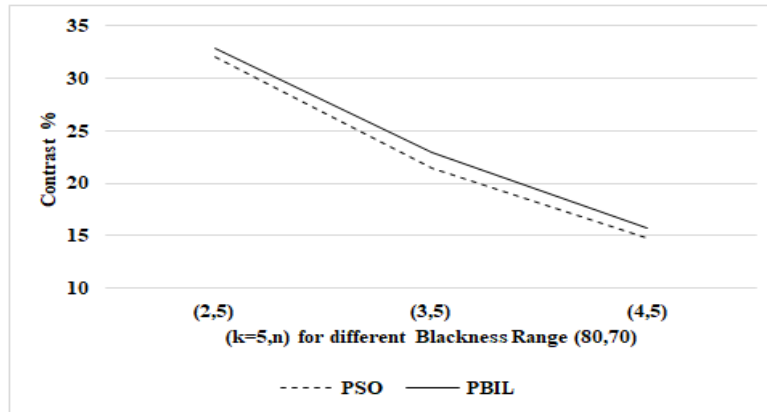


Figure 6 Contrast for PBIL

5 Conclusion

In the VC, a reconstructed image after the process of decryption has encountered a major issue. The quality of the image of such a reconstructed image may not always be similar to the original image owing to the expansion of the pixel. For the purpose of this work, a nature-inspired technique of optimization like the PBIL or the PSO will be applied in order to enhance the image since it is a very interesting application of metaheuristics. The metaheuristic algorithm will enhance the VC shares even before it has been sent to the network and thus there is no necessity for computation at the receiving end. This fitness function can be quite simple and may be applied to the optimization of those problems that are large in terms of dimensions that may produce quality solutions faster. This PBIL was for the purpose of generating a real-valued probability vector that, when sampled, can result in solutions of high quality with higher probability. The primary aim of this method was to improve the decrypted VC image by means of using appropriate fitness functions even before they get transmitted to an end-user for its decryption. Results proved that the PBIL had a higher contrast by about 2.53% for (2, 3), a blackness range (100, 90), by about 2.55% for (2, 4) the blackness range (100, 90), by about 2.61% for (2, 5) the blackness range (100, 90), by about 2.7% for (2, 6) the blackness range (100, 90), by about 2.54% for (2, 7) the blackness range (100, 90), and finally, by about 2.21% for (2, 8) the blackness range (100, 90) on being compared to the PSO respectively.

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